

JEE MAIN CUMULATIVE TEST SERIES



NARAYANA
IIT ACADEMY
INDIA



Sec: JR_*CO-SC(MODEL-B)

Date: 23-06-24

Time: 3 Hrs

Max. Marks: 300

23-06-24_JR.IIT_STAR CO-SC(MODEL-B)_JEE MAIN_CTM-1_SYLLABUS

PHYSICS:

PRESENT WEEK (80%): FRICTION: Contact Force and its components, Dry Friction, Static and Dynamic Friction, Laws of Friction, Angle of Friction & Repose, Rolling Friction

PREVIOUS WEEK (20%): Springs, Cutting & Combination of Springs, constraint equations, Problems in NLM without friction

CHEMISTRY:

PRESENT WEEK (80%): Buffer solution & Buffer capacity, Solubility of sparingly soluble salts and solubility product, Indicator, Acid base titration curves

PREVIOUS WEEK (20%): Ionic Equilibrium-I: Theories of acid & bases – Arrhenius, Bronsted-Lowry and Lewis theory (**Exclude: Comparison of Acidic & Basic strength**), Ionic product of water and pH scale pH calculation involving strong acid, Strong base and their mixtures, pH of weak acids and bases, Mixture of weak acids, Levelling effect, Common ion effect, Dissociation of polyprotic acids, Salts and their Hydrolysis

MATHEMATICS:

PRESENT WEEK (80%): Conditional Identities, Summation of Trigonometric series, Miscellaneous problems on Transformation

PREVIOUS WEEK (20%): TRIGONOMETRY: Measurement of Angles, Trigonometric Ratios & Identities, T- Ratio of Compound angle, T-Ratio of Multiple, sub-multiple angles (Excluding graph, periodicity and extreme values),

T-Ratio of some special angles, Transformation formulae

(Except Summation)

PHYSICS

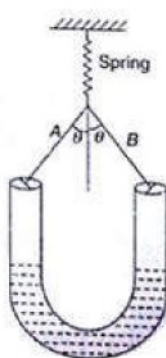
MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

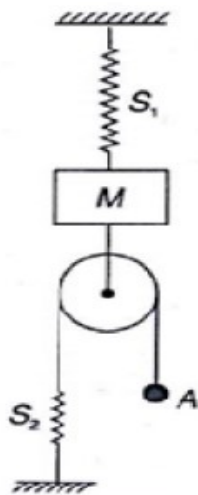
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A U-shaped container has uniform cross sectional area 'S'. It is suspended vertically with the help of a spring and two strings A and B as shown in the figure. The spring and the strings are light. When water of density 'd' is poured slowly in to the container it is observed that the level of water remained unchanged with respect to the ground. The spring constant of spring is

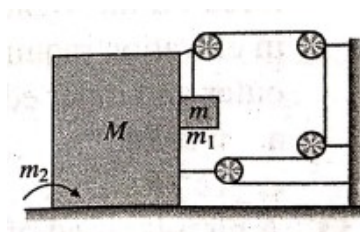


- A) $2Sdg$ B) Sdg C) $Sdg/2$ D) $Sdg/4$
2. In the system shown in the figure the two springs S_1 and S_2 have force constant 'K' each. Initially the system is in equilibrium with spring S_1 stretched and the spring S_2 relaxed. The end A of the string is pulled down slowly through a distance 'L'. The distance by which the block of mass 'M' moves is

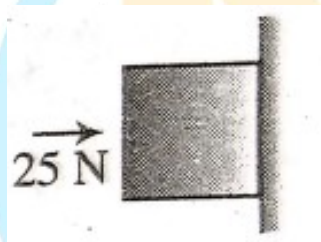


- A) $2L/3$ B) $2L/5$ C) $L/2$ D) $L/5$

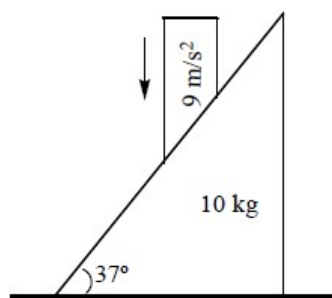
3. Two blocks M and m are arranged as shown in the diagram. The coefficient of friction between the blocks is $\mu_1 = 0.25$ and between the ground and M is $\mu_2 = \frac{1}{3}$. If $M = 8$ kg, then find the value of m so that the system will remain at rest.



- A) $\frac{4}{3}$ kg B) $\frac{8}{9}$ kg C) 1 kg D) $\frac{8}{5}$ kg
4. A given object takes n times more time to slide down 45° rough inclined plane as it takes to slide down a perfectly smooth 45° incline. The coefficient of kinetic friction between the object and the incline is
- A) $\sqrt{\frac{1}{1-n^2}}$ B) $\sqrt{1-\frac{1}{n^2}}$ C) $1-\frac{1}{n^2}$ D) $\frac{1}{2-n^2}$
5. A horizontal force of 25 N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.4. The weight of the block is

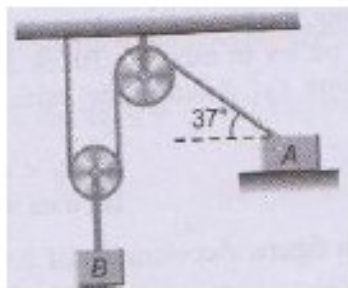


- A) 2.5 N B) 20 N C) 10 N D) 5 N
6. In the system shown in the fig. all surface are smooth. Rod is moved by external agent with acceleration 9 m/s^2 vertically downwards. Force exerted on the rod by the wedge will be

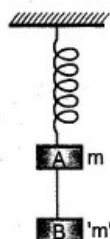


- A) 120 N B) 200 N C) 160 N D) 180 N

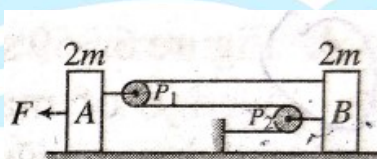
7. In the figure shown the block B moves with the velocity 10 ms^{-1} . The velocity of A in the position shown is [A remains in contact with the floor]



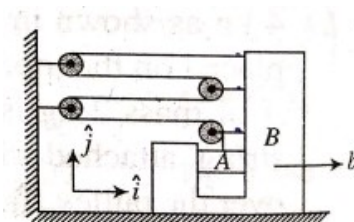
- A) 12.5 ms^{-1} B) 25 ms^{-1} C) 8 ms^{-1} D) 16 ms^{-1}
8. System shown in figure is at equilibrium in vertical plane. Consider the string and spring to be massless. Both the blocks have same mass ' m '. If the string is cut, then just after cutting:



- A) Acceleration of 'A' is ' g ' upwards B) Acceleration of 'A' is ' $2g$ ' upwards
C) Acceleration of 'B' is ' g ' upwards D) Acceleration of 'B' is ' $2g$ ' upwards
9. The acceleration of the block B in Fig, assuming the surfaces and the pulleys P_1 and P_2 are all smooth, is

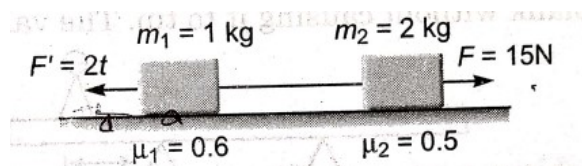


- A) $\frac{F}{4m}$ B) $\frac{3F}{13m}$ C) $\frac{F}{2m}$ D) $\frac{3F}{17m}$
10. If block B moves towards right with acceleration b , find the net acceleration of block A



- A) $b\hat{i} + 4b\hat{j}$ B) $b\hat{i} + b\hat{j}$ C) $b\hat{i} + 2b\hat{j}$ D) None of these

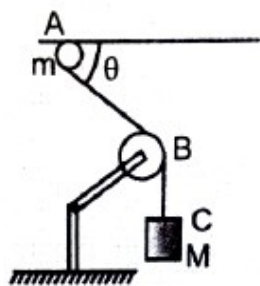
11. Two blocks A and B are separated by some distance and tied by a string as shown in the figure. The force of friction in both the blocks at $t = 2$ s is



- A) $4N(\rightarrow), 5N(\leftarrow)$ B) $2N(\rightarrow), 5N(\leftarrow)$
 C) $0N(\rightarrow), 10N(\leftarrow)$ D) $1N(\leftarrow), 10N(\leftarrow)$
12. Consider the situation shown in figure. Mass of the block and the pulley is m . Assume that string and spring are light, find the elongation of the spring at equilibrium

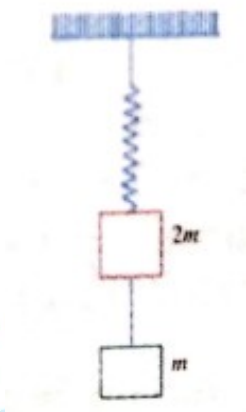


- A) $\frac{2mg}{k}$ B) $\frac{4mg}{k}$ C) $\frac{8mg}{k}$ D) $\frac{3mg}{k}$
13. A smooth ring of mass m can slide on a fixed horizontal rod. A string tied to the ring pass over a fixed pulley B and carries a block C of mass $2m$ as shown below. As the ring starts sliding

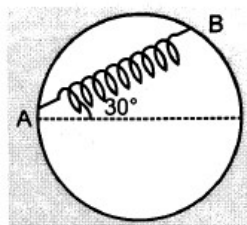


- A) The acceleration of the ring is $\frac{2g \cos \theta}{1 + 2 \cos^2 \theta}$
 B) The acceleration of the block is $\frac{2g}{1 + 2 \cos^2 \theta}$
 C) The tension in the string is $\frac{mg}{1 + 2 \cos^2 \theta}$
 D) The tension in the string is $\frac{mg}{1 + \cos^2 \theta}$

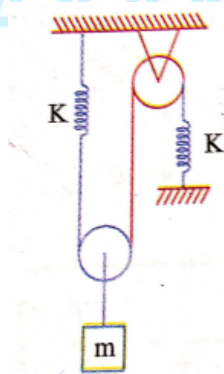
14. The string between blocks of masses 'm' and '2m' is massless and inextensible. The system is suspended by a massless spring as shown. If the string is cut, the magnitudes of accelerations of masses 2m and m (immediately after cutting)



- A) g, g B) $g, \frac{g}{2}$ C) $\frac{g}{2}, g$ D) $\frac{g}{2}, \frac{g}{2}$
15. A bead of mass 'm' is attached to one end of a spring of natural length R & spring constant $K = \frac{(\sqrt{3}+1)mg}{R}$. The other end of the spring is fixed at point A on a smooth vertical ring of radius R as shown. The normal reaction at B just after it is released to move is

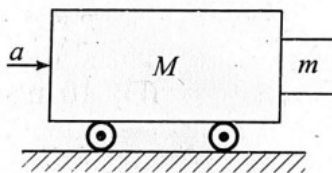


- A) $\sqrt{3}mg$ B) $3\sqrt{3}mg$ C) $\frac{mg}{2}$ D) $\frac{3\sqrt{3}mg}{2}$
16. Block is attached to system of springs. The equivalent spring constant is

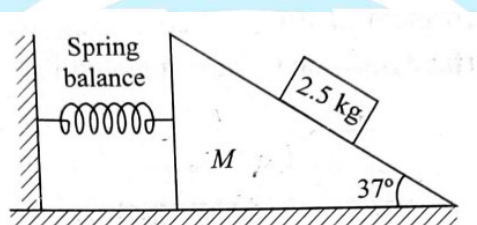


- A) K B) 2K C) 3K D) 4K

17. A spring of constant k cut into '3' equal parts and all the parts connected in parallel, the effective spring constant of the combination is xk , where x is
- A) 3 B) 6 C) 9 D) 12
18. A cart of mass M has a block of mass m in contact with it as shown in the figure. The coefficient of friction between the block and the cart is μ . What is the minimum acceleration of the cart so that the block m does not fall



- A) μg B) g / μ C) μ / g D) $M \mu g / m$
19. Find the reading of spring balance as shown in figure. Assume that mass M is in equilibrium (All surfaces are smooth)



- A) 8 N B) 9 N C) 12 N D) Zero
20. The speed of the block B when the wedges A and C are moving towards each other with speed v and the strings connected to block make an angle $\theta = 53^\circ$ with vertical, as shown in figure is $V_B = v / x$, where x is?
- A) 1 B) 2 C) 3 D) 4

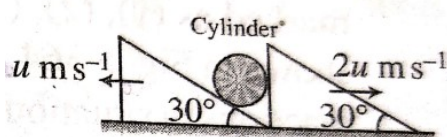
SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

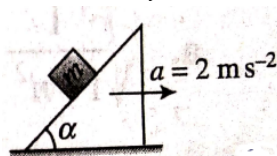
This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only**. Have to Answer any 5 only out of 10 questions and question will be evaluated according to the following marking scheme:

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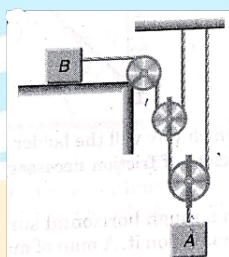
21. A system is shown in fig. Assume that the cylinder remains in contact with the two wedges. Then the velocity of cylinder is $\sqrt{x} \text{ ms}^{-1}$ find x



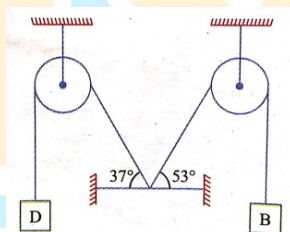
22. A block of mass m is lying on a having inclination angle $\alpha = \tan^{-1}\left(\frac{1}{5}\right)$ wedge is moving with a constant acceleration $a = 2 \text{ ms}^{-2}$. The minimum value of coefficient of friction μ so that m remains stationary w.r.t. wedge is $\frac{x}{y}$. Find $x + y$



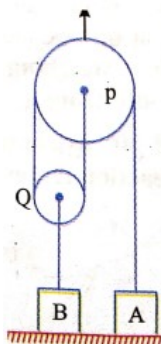
23. Mass of block A is 5 kg. Neglect the mass of the pulley and cords. The block B has a mass of 10 kg. The coefficient of kinetic friction between block B and the surface is $\mu_k = 0.1$. (Take $g = 10 \text{ m/s}^2$). If the acceleration of block A is $\frac{2}{y}$ Find y



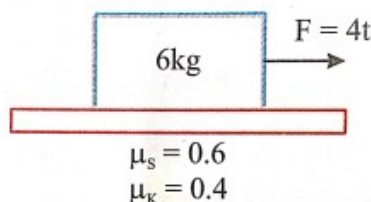
24. A bead C can move freely on a horizontal rod. The bead is connected by blocks B and D by a string as shown in the figure. If the velocity of B is v . The velocity of block D is $4v/x$, then the value of x is



25. Two blocks A and B having masses 1 kg and 4 kg respectively are arranged as shown in the figure. The pulleys P and Q are light and frictionless. All the blocks are resting on a horizontal floor and the pulleys are held such that strings remains just taut. At moment $t = 0$ a force $F = 30t(N)$ starts acting on the pulley P along vertically upward direction as shown in the figure. The time when the blocks A and B loose contact with ground is $4/x$ sec then x is



26. A 6kg block is kept over a rough surface with coefficients of friction $\mu_s = 0.6$ and $\mu_k = 0.4$ as shown in fig. A time varying force $F = 4t$ (F in newton and t in second) is applied on the block as shown. Find the acceleration of block at $t = 5$ sec. ($g = 10 \text{ m/s}^2$)



27. A block of mass 5kg is lying on a rough horizontal surface. The coefficient of static and kinetic friction are 0.3 and 0.1 and $g = 10 \text{ ms}^{-1}$. The frictional force on the block is (in N)
28. A heavy uniform chain lies on horizontal table top. If the coefficient of friction between the chain and the table surface is 0.25, the maximum percentage of the length of the chain that can hang over the edge of the table is
29. A given object taken 3 times as much time to slide down a 45° rough incline as it takes to slide down a perfectly smooth 45° incline. The coefficient of friction between the object and the incline is $\frac{Y}{9}$. Find Y
30. A block slides down a rough inclined plane of slope angle θ with a constant velocity. It is then projected up the same plane with an initial velocity v. The distance travelled by the block up the plane before coming to rest is $\frac{pv^2}{qg \sin \theta}$. Find $p + q$

THE NARAYANA GROUP

CHEMISTRY

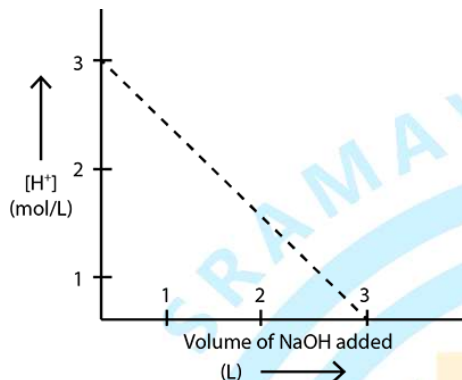
MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

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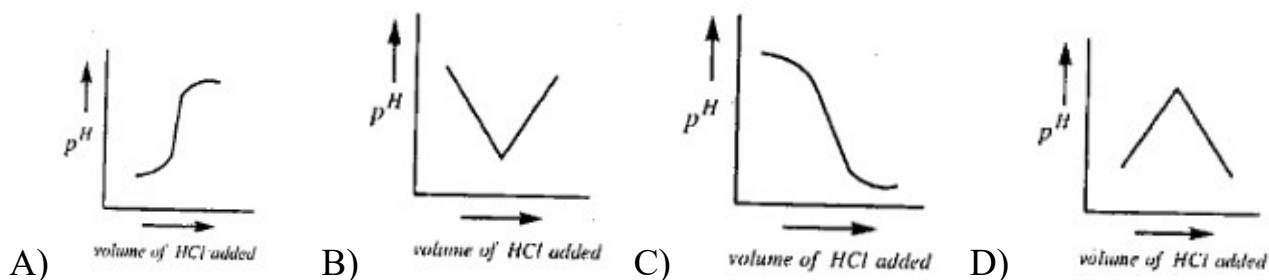
31. 1M NaOH solution was slowly added into 1000 mL of 183.75 g impure H_2SO_4 solution and the following plot was obtained. The Percentage purity of H_2SO_4 sample and slope of the curve respectively are:



- A) 75%, $-\frac{1}{3}$ B) 80%, $-\frac{1}{2}$ C) 80%, -1 D) None of these
32. 1 mL of 10^{-6} is diluted to 10 mL. Thus $[\text{H}_3\text{O}^+]$ in dilute HCl is approximately
- A) 10^{-7} M B) 10^{-14} M C) $2 \times 10^{-7} \text{ M}$ D) 10^{-8} M
33. The pH of 0.1M aqueous solution of the following compounds increases in the order
- A) $\text{NaCl} < \text{NH}_4\text{Cl} < \text{NaCN} < \text{HCl}$ B) $\text{HCl} < \text{NH}_4\text{Cl} < \text{NaCl} < \text{NaCN}$
 C) $\text{NaCN} < \text{NH}_4\text{Cl} < \text{NaCl} < \text{HCl}$ D) $\text{HCl} < \text{NaCl} < \text{NaCN} < \text{NH}_4\text{Cl}$
34. MX is the salt of weak base MOH and weak acid HX. Aqueous solution of MX is
- A) Acidic if $K_a < K_b$ B) Basic, if $K_a < K_b$
 C) Neutral, if $K_a < K_b$ D) All the above
35. Which of the following mixture acts as buffer solution?
- A) 100 mL 0.2 M CH_3COOH + 100 mL 0.1M NaOH
 B) 100 mL 0.1 M CH_3COOH + 100 mL 0.2 M NaOH
 C) 100 mL 0.2 M CH_3COOH + 100 mL of 0.2 M NaOH
 D) None of the above acts as buffer solution due to the presence of strong base NaOH

36. When 0.1 mole of an acid is added to 2 L of a buffer solution, the pH of the buffer decreases by 0.5. The buffer capacity of the solution is
- A) 0.6 B) 0.4 C) 0.2 D) 0.1
37. What is the pH at 25°C of a $4.0 \times 10^{-4}\text{N}$ solution of $\text{Ba}(\text{OH})_2$
- A) 3.4 B) 10.6 C) 3.1 D) 10.9
38. In the titration of acetic acid versus sodium hydroxide, the pH of the solution at equivalence point (when temperature is 25°C) is
- A) about 5.5 B) about 6.5 C) about 7 D) about 8.3
39. When 100 mL of 0.4M CH_3COOH are mixed with 100 mL of 0.2 M NaOH , the $[\text{H}_3\text{O}^+]$ in the solution is approximately $[K_a(\text{CH}_3\text{COOH}) = 1.8 \times 10^{-5}]$
- A) 1.8×10^{-6} B) 1.8×10^{-5} C) 9×10^{-6} D) 9×10^{-5}
40. Given below are two statements:
- Statement (I) : Aqueous solution of ammonium carbonate is basic.
- Statement (II) : Acidic/basic nature of salt solution of a salt of weak acid and weak base depends on K_a and K_b value of acid and the base forming it.
- In the light of the above statements, choose the most appropriate answer from the options given below :
- A) Both Statement I and Statement II are correct
- B) Statement I is correct but Statement II is incorrect
- C) Both Statement I and Statement II are incorrect
- D) Statement I is incorrect but Statement II is correct
41. The solubility of calcium Phosphate in aqueous solution is $y\text{M}$. Then its solubility product is (Where M is mol/L)
- A) $Y^2\text{M}^2$ B) $27 Y^5 \text{M}^5$ C) $108 Y^5 \text{M}^5$ D) $4 Y^5 \text{M}^5$

42. Titration curve if a strong base is titrated with strong acid is

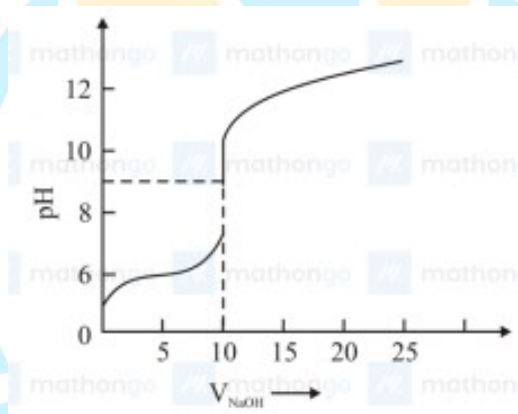


43. K_{sp} of $MX = K_{sp}$ of MX_2 . Which is more soluble?

- A) MX B) MX_2 C) both equal D) K_{sp} values are required

44. The titration curve of weak acid vs. strong base with phenolphthalein as indicator is shown below. The $K_{\text{phenolphthalein}} = 4 \times 10^{-10}$

Given: $\log 2 = 0.3$



The number of following statement/s which is/are correct about phenolphthalein is

- A) It can be used as an indicator for the titration of weak acid with weak base.
 B) It begins to change colour at $\text{pH} = 8.4$
 C) It is a weak organic base
 D) It is colourless in acidic medium

45. If the solubility of Ag_2SO_4 in $10^{-2} \text{MNa}_2\text{SO}_4$ solution be $2 \times 10^{-8} \text{M}$ then K_{sp} of Ag_2SO_4 will be:

- A) 32×10^{-24} B) 16×10^{-18} C) 32×10^{-18} D) 16×10^{-24}

46. Which of the following Aqueous salt solution whose P^H is Less than 7 or it forms an acidic Solution
- A) Ammonium Acetate B) Ammonium Chloride
C) Potassium Chloride D) Sodium Acetate
47. The arrange solubility order of silver chloride in ($K_{sp} = 10^{-10}$)
- i) 0.1 M $AgNO_3$ ii) 0.5 M KCl iii) 0.06 M $BaCl_2$
- A) i>ii>iii B) i>iii>ii C) iii>i>ii D) ii>iii>i
48. Most suitable indicator of the titration of Strong Acid verses Strong Base
- A) Methyl red B) Methyl Yellow
C) Phenol Red D) Methyl Orange
49. Which of the following is a Lewis Acid
- A) H_2O B) NH_3 C) NH_4^+ D) SiF_4
50. Equal volumes of 0.2 M solutions of sodium iodate and cupric chlorate are mixed together. (For cupric iodate $K_{sp} = 7.4 \times 10^{-8}$).
- A) Cupric Iodate is Precipitated because Ionic Product is greater than its K_{SP}
B) Cupric Iodate is Precipitated because Ionic Product is less than its K_{SP}
C) Cupric Iodate is not Precipitated because Ionic Product is equal to its K_{SP}
D) Sodium Chlorate is Precipitated

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

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51. The solubility of $AgCl$ in 0.10 M $BaCl_2$ is $p \times 10^{-10}$, 'p' is ____
- (Given K_{sp} of $AgCl = 10^{-10}$).
52. Calculate the percentage hydrolysis in a 10^{-2} M KCN solution ($k_a = 6.2 \times 10^{-10}$)

53. A solution of saturated CaF_2 is found to contain 4.0×10^{-4} M fluoride ion. The K_{sp} of CaF_2 is $y \times 10^{-12}$, 'y' is ____ (Neglect hydrolysis)
54. A certain buffer solution contains equal concentration of X^- and HX . K_b for X^- is 10^{-10} . Find the pH of buffer
55. The pH of an aqueous solution containing 1M benzoic acid ($\text{pK}_a = 4.20$) and 1M sodium benzoate is 4.5. The volume of benzoic acid solution in 300 mL of this buffer solution is ____ mL.
56. The pH at which $\text{Mg}(\text{OH})_2$ [$K_{sp} = 1 \times 10^{-11}$] begins to precipitate from a solution containing 0.10Mg^{2+} ions is
57. pH of $\text{Ba}(\text{OH})_2$ is 12. If its K_{sp} is $x \times 10^{-y} \text{M}^3$ then the value of $\frac{y}{x}$ is
58. 20 mL of 0.1M NaOH is added to 50 mL of 0.1M acetic acid solution. The pH of the resulting solution is $\times 10^{-2}$. (Nearest integer)
Given : $\text{pK}_a(\text{CH}_3\text{COOH}) = 4.76$, $\log 2 = 0.30$, $\log 3 = 0.48$
59. The minimum pH required to prevent the precipitation of ZnS in a solution containing 0.01M ZnCl_2 saturated with $0.10 \text{M H}_2\text{S}$ is
(Given $K_{sp}(\text{ZnS}) = 10^{-21}$, $K_{a1} \times K_{a2}$ of $\text{H}_2\text{S} = 10^{-20}$)
60. Solubility of $\text{Pb}(\text{OH})_2$ (s) in pure water is 10^{-4}M . How many times the solubility of $\text{Pb}(\text{OH})_2$ will increase in buffer of $\text{pH} = 10$?

MATHEMATICS

MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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61. The value of the sum of the series $\sqrt{1+\frac{1}{1^2}+\frac{1}{2^2}}+\sqrt{1+\frac{1}{2^2}+\frac{1}{3^2}}+\sqrt{1+\frac{1}{3^2}+\frac{1}{4^2}}+\dots$ (2023 terms) is equal to:
- A) $\frac{2022 \times 2020}{2021}$ B) $\frac{2023 \times 2021}{2022}$ C) $\frac{2024 \times 2022}{2023}$ D) $\frac{2025 \times 2023}{2024}$
62. The sum of series $\left(2^{19} + \frac{1}{2^{19}}\right) + 2\left(2^{18} + \frac{1}{2^{18}}\right) + 3\left(2^{17} + \frac{1}{2^{17}}\right) + \dots + 19\left(2 + \frac{1}{2}\right) + 20$ is equal to:
- A) $2^{21} + \frac{1}{2^{19}} - 4$ B) $2^{21} + \frac{1}{2^{18}} - 16$ C) $2^{21} - \frac{1}{2^{19}} + 24$ D) $2^{21} - \frac{1}{2^{18}} + 10$
63. Let $2a = 2 \tan 10^\circ + \tan 50^\circ$, $2b = \tan 20^\circ + \tan 50^\circ$, $2c = 2 \tan 10^\circ + \tan 70^\circ$ and $2d = \tan 20^\circ + \tan 70^\circ$. Which of the following is correct?
- A) $a > b < c > d$ B) $a < b < c < d$ C) $a < b > c > d$ D) $a > b > c > d$
64. Consider two different arithmetic progressions $A_1, A_2, A_3, \dots, A_n$ and $B_1, B_2, B_3, \dots, B_n$ where $1 \leq n \leq 20$ with positive terms such that $\frac{A_n}{B_1} = \frac{\sum_{i=1}^n 2A_i}{\sum_{i=1}^n B_i} = \frac{B_n}{A_1} = 4$. The value of $\frac{B_{10} - B_8}{A_{12} - A_{11}}$ is equal to:
- A) $\frac{1}{13}$ B) $\frac{1}{10}$ C) $\frac{2}{15}$ D) $\frac{4}{7}$
65. Let $A = \{n \in \mathbb{N} : \text{H.C.F}(n, 45) = 1\}$ and $B = \{2k : k \in \{1, 2, \dots, 100\}\}$. Then the sum of all the elements of $A \cap B$ is equal to:
- A) 5264 B) 5100 C) 4280 D) 4470

66. If $\frac{\cos x - \cos \alpha}{\cos x - \cos \beta} = \frac{\sin^2 \alpha \cos \beta}{\sin^2 \beta \cos \alpha}$ then $\tan^2 \frac{\alpha}{2} \tan^2 \frac{\beta}{2} =$
- A) $\cot^2 x$ B) $\cot^2 \frac{x}{2}$ C) $\tan^2 x$ D) $\tan^2 \frac{x}{2}$
67. $\frac{3 + \cot 76^\circ \cot 16^\circ}{\cot 76^\circ + \cot 16^\circ} =$
- A) $\tan 46^\circ$ B) $\cot 76^\circ$ C) $\tan 16^\circ$ D) $\cot 44^\circ$
68. If the inequality $4^x - a2^x - a + 3 \leq 0$, where a is real parameter is satisfied by at least one positive real value of x then the complete set of values of a is equal to:
- A) $(-\infty, 2)$ B) $(1, \infty)$ C) $(-2, 3)$ D) $(2, \infty)$
69. The value of $\cos^6 \frac{\pi}{16} + \cos^6 \frac{3\pi}{16} + \cos^6 \frac{5\pi}{16} + \cos^6 \frac{7\pi}{16}$ is equal to:
- A) $3/2$ B) $3/4$ C) $5/2$ D) $5/4$
70. The value of $2\sqrt{2} \sin 10^\circ \left(\frac{\sec 5^\circ}{2} + \frac{\cos 40^\circ}{\sin 5^\circ} - 2 \sin 35^\circ \right)$ is equal to:
- A) 4 B) 2 C) 1 D) -2
71. The sum of all integer values of K in $[-4, 7]$ for which $\left(\frac{x}{x^2 - 5x + 9} \right)^2 + \left(\frac{x}{x^2 - 5x + 9} \right) + K < 0$ for all real values of x is equal to:
- A) -7 B) 7 C) 9 D) -9
72. Let $R_1 = \{(a, b) \in \mathbb{N} \times \mathbb{N} : |a - b| \leq 13\}$ and $R_2 = \{(a, b) \in \mathbb{N} \times \mathbb{N} : |a - b| \neq 13\}$. Then on $\mathbb{N}$:
- A) Both R_1 and R_2 are equivalence relations
- B) Neither R_1 nor R_2 is an equivalence relation
- C) R_1 is an equivalence relation but R_2 is not
- D) R_2 is an equivalence relation but R_1 is not

73. Let a, b, c be the sides of a triangle where $a \neq b \neq c$ and $\lambda \in R$. If the roots of the equation $x^2 + 2(a+b+c)x + 3\lambda(ab+bc+ca) = 0$ are real, then
- A) $\lambda \in \left(\frac{1}{3}, \frac{5}{3}\right)$ B) $\lambda < \frac{4}{3}$ C) $\lambda > \frac{5}{3}$ D) $\lambda \in \left(\frac{4}{3}, \frac{5}{3}\right)$
74. If $x, y, z, \in R$ & $x^2 + 6y^2 + z^2 - 4xy - 2yz - 2y + 1 = 0$, then $x + y + z =$
- A) 1 B) 2 C) 3 D) 4
75. $[\log_2^3] + [\log_2 4] + \dots + [\log_2 20] =$ where $[.]$ represents greatest integer function
- A) 51 B) 52 C) 53 D) 54
76. A class has 175 students. The following data shows the number of students obtaining one or more subjects. Mathematics 100, Physics 70, Chemistry 40, Mathematics and Physics 30, Mathematics and Chemistry 28, Physics and Chemistry 23, Mathematics, Physics and Chemistry 18. How many students have offered Mathematics alone if every student has opted atleast one subject ?
- A) 35 B) 48 C) 3 D) 4
77. Among the following statements, number of correct statement(S) is/are ?
- a) $n((A \times B) \cup (A \times C)) = 12$, where $A = \{1, 2, 3\}, B = \{3, 4\}, C = \{4, 5, 6\}$
- b) If $P(A) = P(B)$, then $A = B$. [A, B are two sets & P (A) denotes power set of A]
- c) For any two sets A & B $A \& B P(A) \cup P(B) = P(A \cup B)$ (P(A) denotes power set of A)
- d) The Cartesian product $A \times A$ has 9 elements among which are found $(-1, 0)$ & $(0, 1)$ then the set is $A = \{-1, 0, 1\}$
- A) 1 B) 2 C) 3 D) 4
78. Let $x_i (1 \leq i \leq 10)$ be ten observations. If $\sum_{i=1}^{10} (x_i - p) = 3$ and $\sum_{i=1}^{10} (x_i - p)^2 = 9$ where $0 \neq p \in R$, then the multiplicative inverse of standard deviation of these observations is:
- A) $\frac{9}{10}$ B) $\frac{10}{7}$ C) $\frac{10}{9}$ D) $\frac{7}{10}$

79. Suppose a, b denote the distinct real roots of the quadratic polynomial $x^2 + 20x - 2020$ & suppose c, d denote the distinct complex roots of the quadratic polynomial $x^2 - 20x + 2020$. Then the value of $ac(a-c) + ad(a-d) + bc(b-c) + bd(b-d)$ is
- A) 0 B) 8000 C) 8080 D) 16000
80. Let α and β be the roots of equation $x^2 - 6x - 2 = 0$, $a_n = \alpha^n - \beta^n$, for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is equal to
- A) 6 B) -6 C) 3 D) -3

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

81. $\tan^2 \frac{\pi}{16} + \tan^2 \frac{2\pi}{16} + \tan^2 \frac{3\pi}{16} + \tan^2 \frac{4\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{6\pi}{16} + \tan^2 \frac{7\pi}{16} = \underline{\hspace{2cm}}$
82. Let $\alpha, \beta, \gamma > 0$ and $\alpha + \beta + \gamma = \frac{\pi}{2}$ if $p = 7 + \tan \alpha \tan \beta$, $q = 5 + \tan \beta \tan \gamma$, $r = 3 + \tan \gamma \tan \alpha$ then the maximum value of $\frac{1}{\sqrt{3}}(\sqrt{q} + \sqrt{q} + \sqrt{r})$ is
83. The value of $\operatorname{cosec} 10^\circ + \operatorname{cosec} 50^\circ - \operatorname{cosec} 70^\circ$ equals
84. Let $Q(x)$ be a quadratic polynomial with real coefficients such that for all real x the relation, $2(1+Q(x)) = Q(x-1) + Q(x+1)$ holds good. If $Q(0) = 8$ and $Q(2) = 32$. Then the number of integral values of 'x' for which $Q(x)$ gives negative real values is
85. The set of real values of x satisfying $\|x-1|-1\| \leq 1$ is $[a, b]$, $\left| \frac{b}{a} \right|$ is equal to
86. Let the median and the mean deviation about the median of 7 observation 170, 125, 230, 190, 210, a , b be 170 and $\frac{205}{7}$ respectively. Then the mean deviation about the mean of these 7 observations is N . Value of $\frac{N}{10}$ is $\underline{\hspace{2cm}}$

87. Value of x for which the equation $\left(x^{\log_{10} 3}\right)^2 - \left(3^{\log_{10} x}\right) - 2 = 0$ is $a^{\log_b c}$, where $a, b, c \in \mathbb{N}$ & a, b are prime numbers then $a + b$ is ____
88. Let α and β be two real numbers such that $\alpha + \beta = 1$ and $\alpha\beta = -1$. Let $p_n = (\alpha)^n + (\beta)^n$, $p_{n-1} = 11$ and $p_{n+1} = 29$ for some integer $n \geq 1$. Then, the value of p_n^2 is ____
89. The number of real roots of $(6-x)^4 + (8-x)^4 = 16$ is
90. Four different integers form an increasing A.P. one of these numbers is equal to the sum of the squares of the other three numbers. The largest of these numbers is equal to

